

OIS-013 — Operational Invariants

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-012
 - OAS-000 through OAS-031
 - OCS-000 through OCS-032
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Purpose

This specification defines the operational invariants that every conforming implementation SHALL preserve during execution.

The Constitution defines what must always remain true.

Architecture defines how those truths are organized.

Operational Invariants define what execution SHALL never violate.

Optimizations MAY change execution.

They SHALL never violate Operational Invariants.

1. Implementation Objective

The objective of Operational Invariants is to guarantee constitutional correctness under every operational condition.

Execution SHALL preserve:

Identity.

Meaning.

Historical continuity.

Relationship integrity.

Reconstructability.

Architectural intent.

Operational correctness SHALL never supersede constitutional correctness.

2. Definition

An Operational Invariant is a property that SHALL remain true throughout every execution state of a conforming implementation.

Operational state MAY change.

Operational Invariants SHALL not.

3. Identity Invariant

Execution SHALL preserve constitutional identity.

Operational identifiers MAY change.

Canonical identity SHALL remain continuous.

No execution path SHALL silently duplicate or replace constitutional identity.

4. Event Invariant

Every constitutionally significant operational change SHALL generate immutable Events.

Execution SHALL never:

Rewrite Events.

Delete Events.

Reorder constitutional history.

Derived operational state SHALL remain reconstructable.

5. Relationship Invariant

Execution SHALL preserve constitutional relationships.

Temporary operational representations SHALL never redefine constitutional relationships.

Relationship loss SHALL constitute implementation failure.

6. Schema Invariant

Execution SHALL conform to the active constitutional Schema.

Schema validation SHALL precede constitutional modification whenever applicable.

Operational shortcuts SHALL not bypass constitutional compatibility.

7. Traceability Invariant

Every observable constitutional behavior SHALL remain traceable.

Execution SHALL preserve:

Origin.

Reasoning.

Historical lineage.

Applicable specifications.

Operational traceability SHALL remain reconstructable.

8. Recovery Invariant

Recovery SHALL preserve constitutional equivalence.

After recovery,

implementations SHALL reconstruct equivalent:

Identity.

History.

Relationships.

Architectural behavior.

Recovery SHALL not alter constitutional meaning.

9. Federation Invariant

Federated execution SHALL preserve:

Local sovereignty.

Canonical identity.

Independent evolution.

Historical continuity.

Synchronization SHALL never redefine constitutional behavior.

10. Failure Invariant

Operational failure SHALL never justify constitutional corruption.

Implementations MAY suspend execution.

They SHALL preserve constitutional continuity.

Failure SHALL remain observable.

11. Verification Invariant

Every Operational Invariant SHALL be independently verifiable.

Verification SHALL produce explicit evidence demonstrating preservation under:

Normal execution.

Recovery.

Migration.

Federation.

Failure.

12. Operational Invariant Test

Operational Invariants satisfy this specification when every execution path, regardless of infrastructure, optimization strategy, failure mode, or deployment architecture, preserves equivalent constitutional behavior without violating any constitutional or architectural invariant.

Conformance

An implementation conforms to this specification when every operational state transition preserves constitutional identity, architectural intent, historical continuity, semantic integrity, and reconstructability independently of execution technology.

Implementation Principle

Execution is temporary.

State is temporary.

Infrastructure is temporary.

Operational Invariants are not.

They exist so that every optimization,

every recovery,

every migration,

and every future implementation remains free to change,
while constitutional truth remains untouched.

End of Specification OIS-013